Mock Regional

11 problems, 5 hours, 1 computer — then the debrief

Learning goals

- Run a full regional simulation under real rules
- Practise triage, the problem board, and keyboard scheduling
- Recognise every technique from Lectures 2–9 in disguise
- Debrief systematically: what cost time, what cost penalty, what to train next

Rules for the mock

- 5 hours, teams of 3, **one computer**, printed notebook allowed
- Standard scoring: solved, then penalty (20 min per rejected try)
- Scoreboard frozen for the last hour
- No internet except the judge; clarifications via the coach
- Problems in alphabetical order — **not** sorted by difficulty

The set at a glance (no spoilers)

	Title		Title
A	Odometer Without Fours	G	Palindrome Banner
B	Arrow Grid	H	Rest Stops
C	Toll Vouchers	I	Coprime Committees
D	Lab Bookings	J	Fence Lattice
E	Double Coverage	K	Forbidden Jingle
F	Gears		

- Statements are summarised in the notes; hand them out printed

Recommended first 20 minutes

- Split: A–D, E–H, I–K (+ template typing)
- Label E / M / H with one-line ideas
- First keyboard: whichever **E** is found first
- Expectation: 2 easy, 1 easy-medium, 5 medium, 2 medium-hard, 1 hard

Debrief

Solutions and what each problem was testing

Intended difficulty and topics

	Difficulty	Topic (lecture)
A	Easy	base conversion (2)
E	Easy	1D sweep (6)
B	Easy-medium	simulation + cycle detection (2)
C	Medium	Dijkstra on layered states (4)
D	Medium	Moore–Hodgson greedy (3)
F	Medium	general CRT (7)
G	Medium	prefix function / palindromic suffix (9)
J	Medium	convex hull + Pick (8)
H	Medium	DP + monotonic deque (5)

A — Odometer Without Fours

- Display skips every number containing digit 4; given the display, find the true distance
- Displayed numbers = **base 9** written with digits {0,1,2,3,5,…,9}
- Map digits 5–9 → 4–8, read in base 9

```
for (char ch : d) { int x = ch - '0'; if (x > 4) x--; v = v * 9 + x; }
```

- Trap: answers reach ≈ $1.35 \cdot 10^{18}$ (19-digit display) → `long long`, never `int`

E — Double Coverage

- n half-open intervals, find total length covered by **≥ K** of them
- Events `(l, +1)`, `(r, −1)`; sort; add `x_next − x` while count ≥ K
- Order at equal x is irrelevant here (zero-length gaps) — know *why*
- Trap: "at least K" vs "exactly K"; coordinates in $[-10^9, 10^9]$ ⇒ length up to $2 \cdot 10^9$ → `long long`

B — Arrow Grid

- Robot follows the arrow in its cell; leaving the grid = stop forever
- $K \le 10^{18}$ steps ⇒ cycle detection on **cells** (the direction is a function of the cell)
- `first[cell]`; on repeat jump with `(K - t) mod len`
- Trap: the "stopped" state must be handled before cycle logic

C — Toll Vouchers

- ≤ 10 vouchers, each makes one road free; cheapest route 1 → n
- State `(city, vouchers used)` : (k+1)·n ≈ $1.1 \cdot 10^6$ states, Dijkstra
- Answer: **min over all layers** (using fewer vouchers is allowed)
- Trap: `int` distances (tolls up to 10^9 × path length)

D — Lab Bookings

- Experiments with durations and deadlines; maximise the number completed
- Sort by deadline, push durations into a max-heap, drop the longest when late (Moore–Hodgson)
- Trap: dropping the *current* experiment instead of the longest
- Proof sketch expected in the debrief (Lecture 3)

F — Gears

- Gear i: m_i teeth, mark at tooth r_i, advances 1 tooth/s; first t with all marks at 0
- `t ≡ −r_i (mod m_i)` for all i — moduli **not** coprime
- General CRT merge with `__int128`; inconsistent ⇒ `never`
- Trap: `t = 0` if all marks are already at 0; normalise `−r_i mod m_i`

G — Palindrome Banner

- Append the fewest letters to the end of s to make a palindrome
- Keep the **longest palindromic suffix**; append reverse of the rest
- Longest palindromic suffix of s = prefix function of `rev(s) + '#' + s`, last value
- $n \le 10^6$: O(n). Trap: forgetting the separator

H — Rest Stops

- Rest at checkpoints with costs; gaps between rests ≤ k; start/end free
- `dp[i] = c_i + min(dp[i−k..i−1])`, sliding-window minimum
- O(n) with a monotonic deque (a segment tree also passes, but is more code)
- Trap: the finish n+1 is a "free rest" that must also respect the gap

J — Fence Lattice

- Fence = convex hull of trees; count lattice points **strictly inside**
- Hull (monotone chain), then Pick: `I = (2A − B + 2) / 2`, `B = Σ gcd`
- Collinear hull points do not change B (gcd counts them anyway)
- Trap: all trees collinear ⇒ answer 0 (no area); `long long` area

I — Coprime Committees

- Count non-empty subsets with gcd 1, values ≤ 10^6
- `F(d) = 2^{c_d} − 1`; exact: `G(d) = F(d) − Σ_{k≥2} G(kd)`, d from V down
- O(V log V) harmonic loops
- Trap: modular subtraction negativity; precompute powers of 2

K — Forbidden Jingle

- Strings of length $n \le 10^{18}$ over σ letters avoiding p ($|p| \le 30$)
- KMP automaton (Lecture 9) → transition matrix `T` ($m \times m$)
- Answer = $\Sigma_j (T^n)[0][j]$; $O(m^3 \log n) \approx 1.6 \cdot 10^6$ operations
- Trap: rows for the pattern-complete state must be **absent**, not absorbing

Expected scoreboard

Solved	Typical team
3	A, E, B — fundamentals in place
5–6	+ C, D or G, J — solid regional team
8–9	+ F, H, I — medal contention
10–11	+ K — regional winner

Debrief template (fill in as a team)

	Read at	Label	Started	AC at	Tries	Time lost on	Root cause
A							
…							

- Root cause ∈ {misread, wrong idea, implementation bug, overflow, TLE, format, idle machine}

Debrief questions

1. Which problem did we solve **latest relative to the field**? Why?
2. How long was the machine idle or used for thinking?
3. Which WA could the 60-second checklist have prevented?
4. Which unsolved problem was closest? What was missing — idea or code?
5. What goes into the notebook / training plan this week?

Upsolving rule

- Every problem: re-solve alone within one week
- Unsolved medium: write the solution, then **explain it** to a teammate
- Add a stress test for every WA you got
- Update the problem board template with what you learned

Summary

- The mock exercises every lecture of this track under contest pressure
- Most points are in the medium band: C, D, F, G, H, J
- The debrief matters more than the score
- Next: National Championship Prep — training systems, advanced data structures, DP optimisations